HS Science Instructional Resource Recommendation



PORTLAND
Public Schools

6/10/25

Agenda

Background and Purpose

Selection Process

Recommendation

PD Plan



Background and Purpose

Oregon requires all public school districts to select and adopt high-quality, research-based materials for instruction.

PPS is engaged in this work during this school year to comply with state requirements for science adoption for 2024-2031. PPS has already utilized an extension to complete the adoption process. Process was conducted in alignment with the Instructional Materials Selection Board Policy.

Board Goals:

- Persistent and Predictable Gaps in Graduation Rates among groups of students
- Persistent achievement gaps in post-secondary readiness (regarding interest and success in advanced coursework through successful completion of core academic courses)

Common HQIM

Adopting High-Quality Instructional Materials (HQIM) gives educators aligned, evidence-based science resources. This allows teachers to focus on instruction, including differentiation and extension, rather than searching for or constructing lesson materials from various books and websites.

Common high school science materials support a district-wide focus on rigorous, engaging, and culturally relevant instruction through ongoing professional development and reflection on student work.

High-quality shared materials ensure all students access rigorous, coherent, grade-level, and culturally relevant science texts and tasks.

It is important to recognize that no purchased set of instructional materials will ever provide the full range of support that is needed for daily classroom instruction, differentiation, and extension. Current research tells us that it is the pedagogical practices and moves that teachers make that can have the greatest impact on student success (Hattie, 2008).

Aligned Curriculum is a baseline that:

- Contains high-quality instructional materials that promote equity
- Provides structure and guidance for novice teachers
- Defines grade-level work clearly
- Serves as a common anchor for conversations about rigor and expectations
- Uses research-based practices

Selection Timeline

HS Science Curriculum Adoption Advisory
Grounded in research based practices, the committee evaluates curriculum materials.

Field Test Training
Training provided for participating teachers. Teachers and student data collected

Fall 2024

Dec. 2024

Feb. 2025

Spring 2025

Advisory Field Test Recommendations
Committee evaluations were analyzed with consensus determining materials for field testing

Recommendation to the School Board

Implementation Timeline

Instructional Resource
With approval from the board, contracts are put in place to order text materials and online resources. Teachers can get demo account access.

Ongoing professional learning
Synchronous and asynchronous professional learning opportunities available to teachers

Summer 2025

Aug. 21, 2025

September -June 2026

Spring 2026

Professional Learning
Core HS Science Teachers will receive professional learning for the new curriculum resources and begin unpacking and planning

Implementation Monitoring
Possible activities include: learning walks, feedback forms, open office hours, etc.

Advisory Committee

Fall 2024

The advisory committee completed a rigorous review of HS Science materials previously approved by the Oregon Department of Education as well as Open Source materials aligned to the Next Generation Science Standards/Oregon Science Standards in order to recommend materials for field testing.

The leadership team was composed of PEC staff as well as HS Administrators.

The Teacher Advisory was composed of teachers from every comprehensive high school and 2 Options schools.

Field Test Materials

For Core NGSS Science Courses

Physics:
OpenSciEd- Carolina Version
HS Science Dimensions

Chemistry:
OpenSciEd- Carolina Version
HS Science Dimensions

Biology:
SAVVAS Miller and Levine Biology
HS Science Dimensions

Field Test Locations:

ACCESS Academy
Benson Polytechnic HS
Cleveland HS
Franklin HS
Grant HS
Ida B. Wells HS
Jefferson HS
McDaniel HS
Roosevelt HS
Metropolitan Learning Center

Criteria for the Review and Adoption of Instructional Materials

Field Test Results

HMH Science Dimensions

There was significant support for HMH in Physics and Biology, and agreement to use HMH Chemistry in order to have stronger coherence and alignment across high school science.

Curriculum Resource Coherence
-Creates Consistency That Makes Learning Easier
-Facilitates Teacher Collaboration & Support

Features of HMH Science Dimensions:

- Aligned to Next Generation Science Standards
- Emphasis on Engineering
- Phenomena-based
- Claims, Evidence, Reasoning structure
- Student Centered: Contextual learning, problem solving, actively doing science, text support

Instructional Materials Costs:

Subject	Proposed Materials	Estimated Cost
Biology	HMH Science Dimensions Biology	793,000.00
Chemistry	HMH Science Dimension Physics	800,000.00
Physics	HMH Science Dimensions Chemistry	780,000.00

Implementation Plan:

- Professional Learning to support pedagogical philosophy of NGSS, structure and components of HMH Science Dimensions, and increasing capacity to meet student needs.
  - District-Wide professional learning
    - In-Person Teacher Professional Development
    - Monthly Remote Learning Sessions
    - Administrator and Coach Professional Learning Sessions
  - In Building PLC time
  - Asynchronous PD contained in HMH Science Dimensions
  - Ongoing support as needed from OTL and OTIS.
- Feedback
  - Regular Feedback via survey form
- Monitoring
  - Classroom visits
    - Success Metrics (aligned to PPS Instructional Framework)

Thank you!

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